

LABORATORY 01 – BASE PLATFORM

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COMPUTER NETWORKS

GROUP 01

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OBJETIVE

Initially, understand the operation and internal behavior of operating systems. In addition, acquire concepts related to server infrastructure. Finally, practice, learn and remember how to install and configure some operating systems using virtualization software like VMware.

MATERIALS

- Computers
- Internet access to research and connect virtual machines
- Virtualization software (VMware)

INTRODUCTION

Internet networks are present all over the world, these are the backbone of modern connection technology that allows the transfer of data, manipulation of devices, seamless communication and more applications. Besides, companies are using complex IT infrastructure, such as cloud services, servers, servers as a service, and complements like containers. For the last couple mentioned, both are running on an operating system, also, in some clouds you need to set up an OS to guarantee correct functionality in the infrastructure.

In this way, we can manage the network structures. For this purpose, we need to learn how to use some Unix base systems and Windows Server for special cases. Obviously, we have to install and configure them in the best way to make them useful, including how they work and distributed. Which is something we are going to do in this lab through virtualization software like VMware, creating some virtual machines to do all the installation and working process and learn about all operating systems.

THEORETICAL FRAMEWORK

Hypervisors

A hypervisor, also known as a *Virtual Machine Monitor* (VMM), is software or hardware that creates and manages virtual machines by stripping hardware resources and allocating them to the VM's. This allows multiple operating systems to run on the same physical hardware independently. This action makes the hardware the host to the VM's called guest.

How hypervisors are classified

Hypervisors are classified into two main types:

- **Bare Metal Hypervisors:**
 - Run directly on the physical hardware without requiring a host operating system.
 - Examples: VMware ESXi, Microsoft Hyper-V, Xen.
- **Hosted Hypervisors:**
 - Run on over a host operating system and rely on it for resource management.
 - Examples: VMware Workstation, Oracle VirtualBox.

Characteristics of hypervisors

The characteristics vary according to the types, although they have some in common:

- **Bare Metal Hypervisors:**
 - High performance and efficiency if they run directly on hardware.
 - Suitable for enterprise virtualization and large scale deployments.
- **Hosted Hypervisors:**
 - Easier to set up and use, suitable for development and testing environments.
 - Slightly lower performance compared to Type 1 due to the reliance on the host OS.
- **Common Characteristics:**
 - Isolation between virtual machines.
 - Allocation of resources such as CPU, RAM, storage and more.
 - Support for snapshots and live migration (depending on the hypervisor).

Architecture of hypervisors

Each hypervisor has an internal structure defined according to its characteristics and capabilities.

- **Bare Metal Architecture:**
 - Has interfaces directly in the physical hardware.
 - Contains a management interface to create and manage VMs.
- **Hosted Architecture:**
 - Operates on a host operating system that interacts with the hardware.
 - The hypervisor runs as an application on the host OS.
 - VMs are created and managed through the hypervisor software.

Cloud computing

Refers to the delivery of computing services like servers, storage, databases, networking, software, etc, over the the cloud (internet infrastructure) paying only for the amount of resources used. It eliminates the need for owning physical hardware, allowing organizations to scale resources as needed.

Hypervisors in cloud computing

As with regular virtualization, the cloud has essential supervisors for creating customized environments for each type of client. This includes resource management for the creation of virtual machines in a single server-focused service. It also allows for scalability and flexibility for a company or client.

Difference between a physical server and a cloud-based server

- **Physical Server Costs:**
 - High upfront investment for hardware, installation, and maintenance.
 - Additional costs for power, cooling, and IT staff.
 - Suitable for organizations with predictable workloads.
- **Cloud-Based Server Costs:**
 - Operates on a subscription or pay as you use model.
 - No upfront hardware costs, expenses are operational rather than capital.

- Ideal for dynamic workloads and scalability.

Containers

Containers are lightweight, portable environments that package an application and its dependencies together. Their architecture includes the **Host OS**, which serves as the foundation; the **Container Runtime** (like Docker or containerd), which manages the containers; **Container Images**, which are pre-built, read-only templates containing all necessary dependencies; and **Containers**, which are isolated instances running applications based on these images.

Similarities and differences between virtual machines and containers

Virtual machines and containers both enable isolated environments for running applications, but they differ in efficiency and use cases. VMs provide full isolation by running a dedicated operating system for each instance, which leads to higher resource usage and slower startup times compared to containers. In contrast, containers achieve lightweight isolation by sharing the host OS kernel, making them more resource-efficient and faster to deploy. While VMs are ideal for running multiple OS environments, containers excel in portability and are better suited for microservices and application scaling.

GENERAL USE OF VMWARE

When we start VMware, we see that there are multiple options and things to do. In this way, we can create the virtual machines that we will use in the laboratory. The process is the same or very similar for each one, since both the amount of resources and the specifications are the same. Thus, we can summarize the process of creating the machine as the following sequence.



New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

C:\RECO\en-us_windows_server_2022_x64_dvd_620

Browse...

Windows Server 2022 detected.
This operating system will use Easy Install. [What's this?](#)

☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

New Virtual Machine Wizard

Choose the Virtual Machine Hardware Compatibility

Which hardware features are needed for this virtual machine?

Virtual machine hardware compatibility

Hardware compatibility: Workstation 17.5 or later

Compatible with: ☒ ESX Server

Compatible products:

Fusion 13.5 or later
Workstation 17.5 or later

Limitations:

128 GB memory
32 processors
10 network adapters
8 TB disk size
8 GB shared graphics memory

Help < Back Next > Cancel

New Virtual Machine Wizard

Firmware Type

What kind of boot device should this virtual machine have?

Firmware type

☐ BIOS

☒ UEFI

☐ Secure Boot

< Back Next > Cancel

New Virtual Machine Wizard

Processor Configuration

Specify the number of processors for this virtual machine.

Processors

Number of processors: 2

Number of cores per processor: 1

Total processor cores: 2

Help < Back Next > Cancel

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

New Virtual Machine Wizard

Memory for the Virtual Machine
How much memory would you like to use for this virtual machine?

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Maximum recommended memory:
27.7 GB

Recommended memory:
384 MB

Guest OS recommended minimum:
32 MB

Help < Back Next > Cancel

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

Maximum disk size (GB):

Recommended size for Other Linux 2.4.x kernel 64-bit: 8 GB

☐ Allocate all disk space now.
Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

New Virtual Machine Wizard

Network Type
What type of network do you want to add?

Network connection

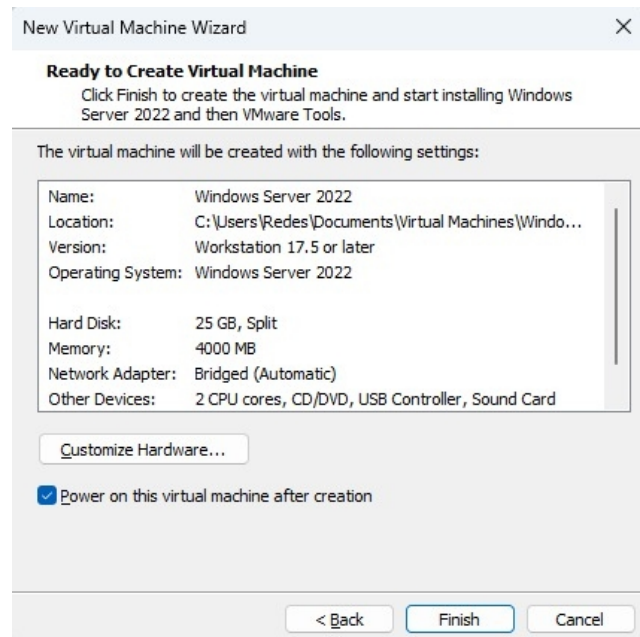
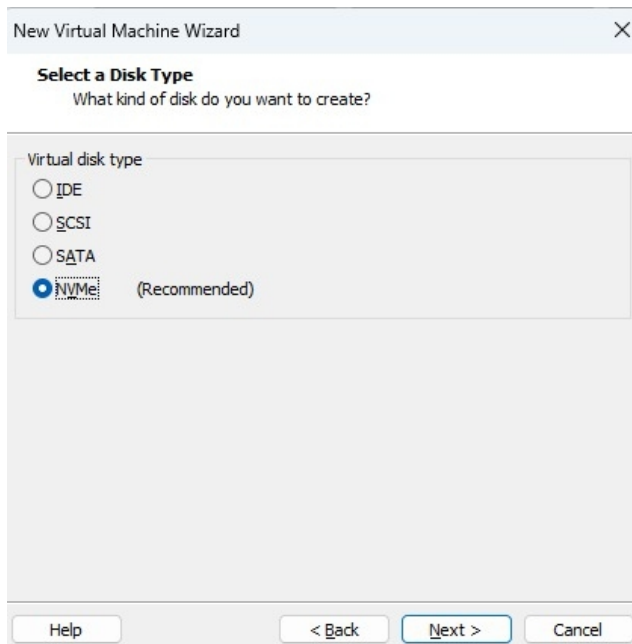
☒ Use bridged networking
Give the guest operating system direct access to an external Ethernet network. The guest must have its own IP address on the external network.

☐ Use network address translation (NAT)
Give the guest operating system access to the host computer's dial-up or external Ethernet network connection using the host's IP address.

☐ Use host-only networking
Connect the guest operating system to a private virtual network on the host computer.

☐ Do not use a network connection

Help < Back Next > Cancel



As we can see, we will use as standard; 4GB of RAM, 10GB of storage (with the exception of Windows Server with GUI that uses 20GB), NVMe formatted hard drive, UEFI boot mode, network card in bridge mode, 2 processor cores and 1 thread for each processor.

With this general configuration for each machine, we can move on to the creation of the required ones for the laboratory.

PRACTICE FRAME

UNIX-BASED SERVER SET UP

Slackware installation

Initially, we are asked to install the Slackware Linux distribution, so that the installation is minimal and without interfaces. For this, we carry out the same process that we just carried out previously for the creation of the virtual machine, assigning the same characteristics as the other machines. Then the system installer will start, where it starts with the first basic configurations, such as the keyboard, partitions and others.



Setting up root Linux partition.

Select Linux installation partition:
Please select a partition from the following list to use for your root (/) Linux partition.

/dev/sda2	Linux 5.06
---	(done adding partitions, continue with setup)
---	(done adding partitions, continue with setup)
---	(done adding partitions, continue with setup)
---	(done adding partitions, continue with setup)
↓(+)	83%

< Select >

<Continue>

Do you want to format Linux partition /dev/sda2?

FORMAT PARTITION /dev/sda2

If this partition has not been formatted, you should format it.
NOTE: This will erase all data on it. Would you like to format this partition?

Format	Quick format with no bad block checking
Check	Slow format that checks for bad blocks
No	No, do not format this partition

< OK >

<Cancel>

Partition /dev/sda2 will be formatted.

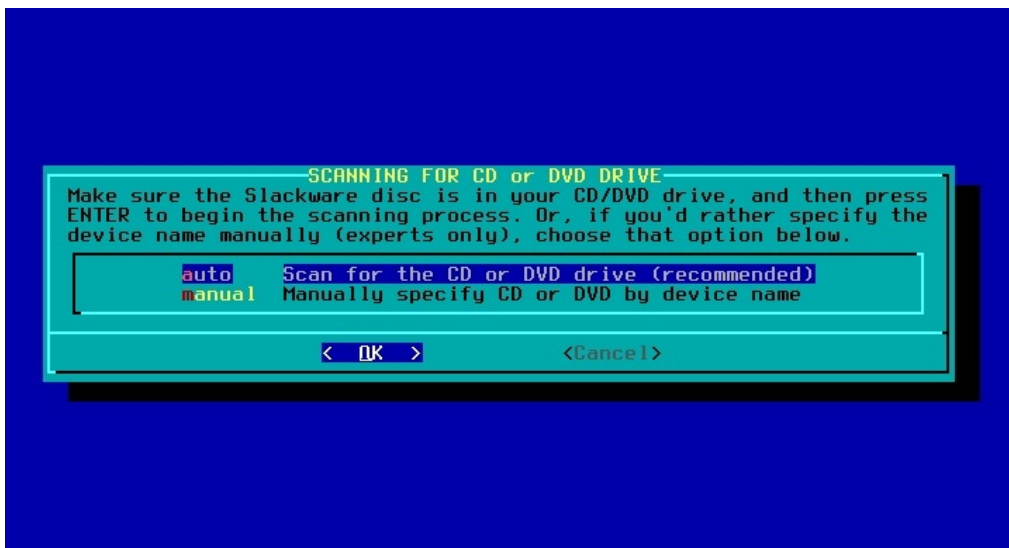
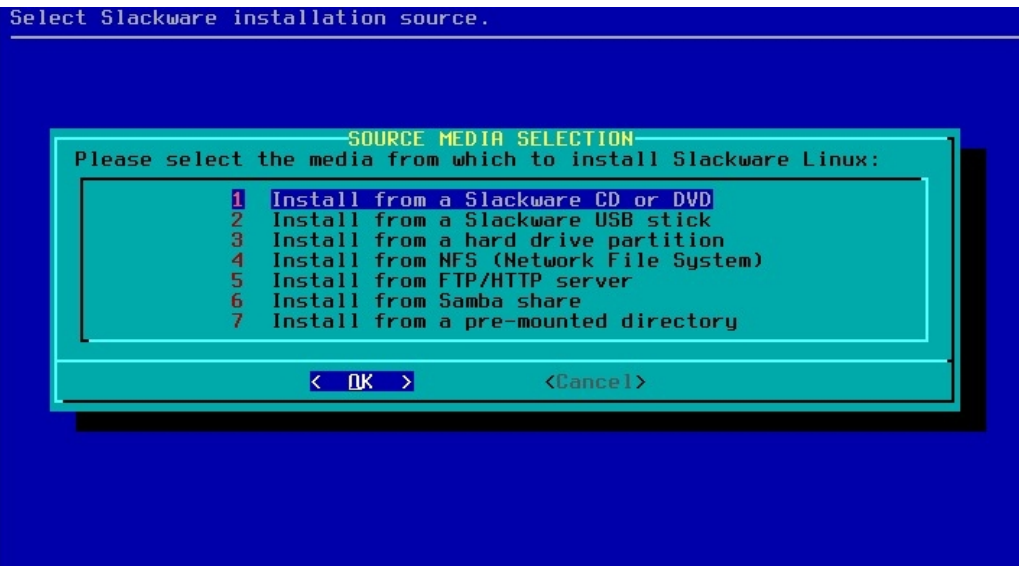
SELECT FILESYSTEM FOR /dev/sda2

Please select the type of filesystem to use for the specified device. Here are descriptions of the available filesystems: Ext2 is the traditional Linux file system and is fast and stable. Ext3 is the journaling version of the Ext2 filesystem. Ext4 is the successor to the ext3 filesystem. Btrfs is a B-tree copy-on-write filesystem. F2FS is a Flash-Friendly File System. JFS is IBM's Journaled Filesystem, currently used in IBM enterprise servers. ReiserFS is a journaling filesystem that stores all files and filenames in a balanced tree structure. XFS is SGI's journaling filesystem that originated on IRIX.

ext2	Standard Linux Ext2 Filesystem
ext3	Ext3 Journaling Filesystem
ext4	Ext4 Journaling Filesystem
jfs	IBM's Journaled Filesystem
reiserfs	ReiserFS Journaling Filesystem
btrfs	Btrfs Copy-on-Write B-tree Filesystem
f2fs	Flash-Friendly File System
xfs	SGI's Journaling Filesystem

< OK >

<Cancel>



With the process shown, we can highlight a couple of points. The first one, the disk was partitioned so that it had a swap in case it was required and that the rest was left in EXT4 format because it is more modern and efficient. Then, we were given the option to go ahead and install packages, where the minimum and necessary ones were chosen so that the system would work correctly, as seen in the following process.

Selecting software to install.

PACKAGE SERIES SELECTION

Now it's time to select which general categories of software to install on your system. Use the spacebar to select or unselect the software you wish to install. You can use the up and down arrows to see all the possible choices. Recommended choices have been preselected. Press the ENTER key when you are finished.

[*]	A	Base Linux system
[*]	AP	Various Applications that do not need X
[]	D	Program Development (C, C++, Lisp, Perl, etc.)
[]	E	GNU Emacs
[]	F	FAQ lists, HOWTO documentation
[]	K	Linux kernel source
[]	KDE	The KDE Plasma Desktop
[*]	L	System Libraries (needed by KDE, GNOME, X, and more)
[*]	N	Networking (TCP/IP, UUCP, Mail, News)

↓(+)

68%

< OK >

<Cancel>

The A (base) series contains the kernel and main system utilities.

SELECT PROMPTING MODE

Now you must select the type of prompts you'd like to see during the installation process. If you have the drive space, the 'full' option is quick, easy, and by far the most foolproof choice. The 'newbie' mode provides the most information but is much more time-consuming (presenting the packages one by one) than the menu-based choices. Otherwise, you can pick packages from menus using 'expert' or 'menu' mode. Which type of prompting would you like to use?

full	Install everything (15+ GB of software, RECOMMENDED!)
terse	Like 'full', but display one line per package during install
menu	Choose individual packages from interactive menus
expert	This is actually the same as the "menu" option
newbie	Use verbose prompting (the X series takes one year)
custom	Use custom tagfiles in the package directories
tagpath	Use tagfiles in the subdirectories of a custom path

↓(+)

87%

< OK >

<Cancel>

Installing package gawk-5.1.1-i586-1 [ADD]

gawk (pattern scanning and processing language)

Gawk is the GNU Project's implementation of the AWK programming language. It conforms to the definition of the language in the POSIX 1003.2 Command Language And Utilities Standard. This version in turn is based on the description in The AWK Programming Language, by Aho, Kernighan, and Weinberger, with the additional features found in the System V Release 4 version of UNIX awk. Gawk also provides more recent Bell Labs awk extensions, and some GNU-specific extensions.

Size: Compressed: 1.1M, uncompressed: 3.1M.

MAKE USB FLASH BOOT

If your computer supports booting from a USB device, it is recommended that you make a USB boot stick for your system at this time. It will boot your computer straight into the root filesystem on /dev/sda2.

Please insert a USB flash memory stick and then press ENTER to create a boot stick.

WARNING! The existing contents of the USB stick will be erased.

Create	Make a USB Linux boot stick
Skip	Skip making a USB boot stick

< OK >

<Cancel>

INSTALL LILO

LILO (Linux Loader) is a generic boot loader. There's a simple installation which tries to automatically set up LILO to boot Linux (also Windows if found). For more advanced users, the expert option offers more control over the installation process. Since LILO does not work in all cases (and can damage partitions if incorrectly installed), there's the third (safe) option, which is to skip installing LILO for now. You can always install it later with the 'liloconfig' command. Which option would you like?

Simple	Try to install LILO automatically
expert	Use expert lilo.conf setup menu
skip	Do not install LILO

< OK >

<Cancel>

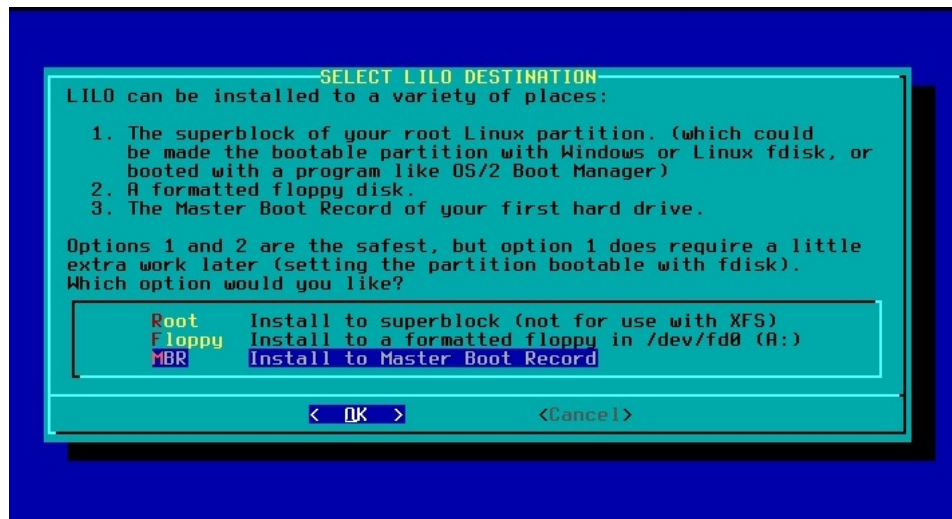
CONFIGURE LILO TO USE FRAME BUFFER CONSOLE?

Looking at /proc/devices, it seems your kernel has support for the VESA frame buffer console. If we enable this in /etc/lilo.conf, it will allow more rows and columns of text on the screen and give you a cool penguin logo at boot time. However, the frame buffer text console is slower than a standard text console. In addition, not every video card or monitor supports all of these video modes, and some X drivers could be confused by them. Would you like to use the frame buffer console, or the standard Linux console?

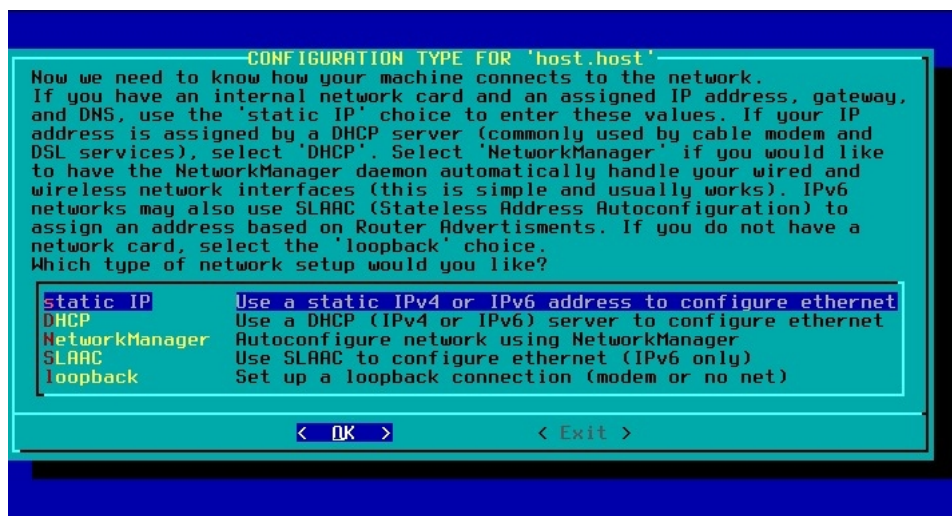
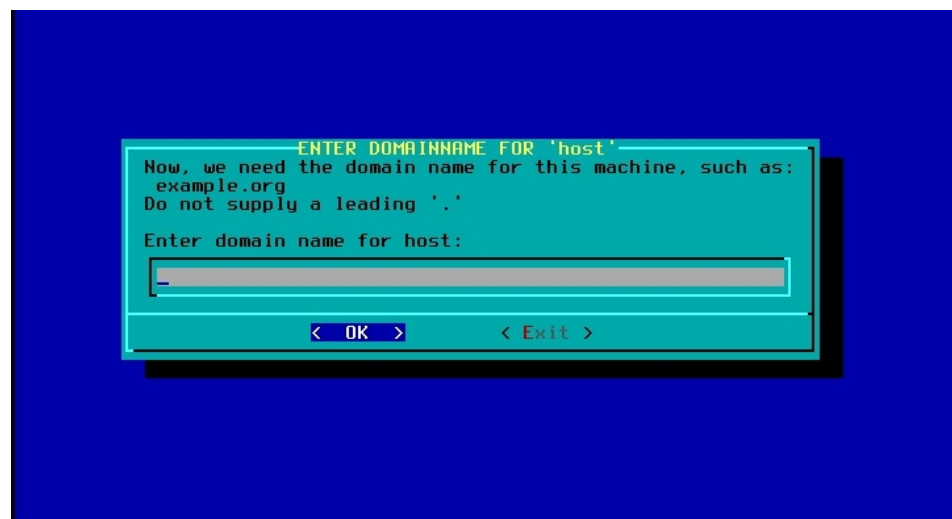
standard	Use the standard Linux console (the safe choice)
ask	Ask about using VESA modes at boot; time out to standard
640x480x64k	Frame buffer console, 640x480x64k
800x600x64k	Frame buffer console, 800x600x64k
1024x768x64k	Frame buffer console, 1024x768x64k
640x480x32k	Frame buffer console, 640x480x32k
800x600x32k	Frame buffer console, 800x600x32k
1024x768x32k	Frame buffer console, 1024x768x32k
640x480x256	Frame buffer console, 640x480x256
800x600x256	Frame buffer console, 800x600x256
1024x768x256	Frame buffer console, 1024x768x256

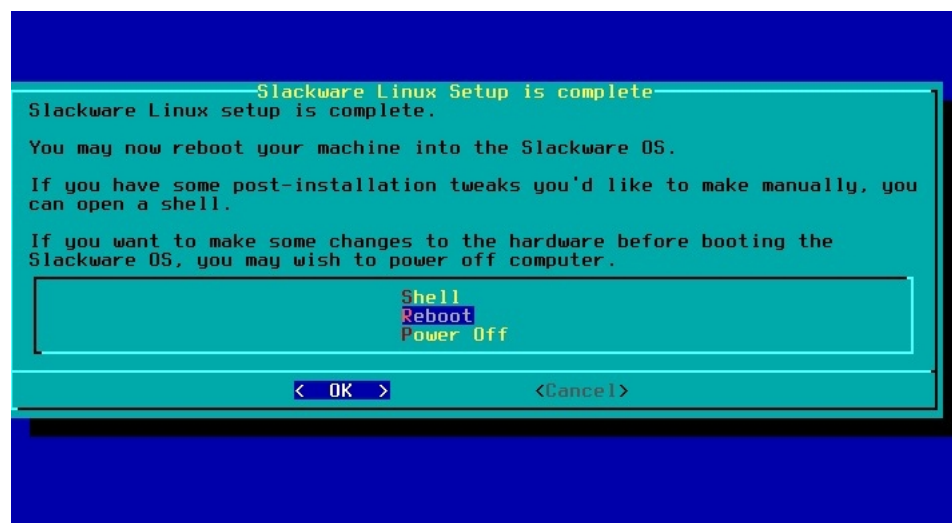
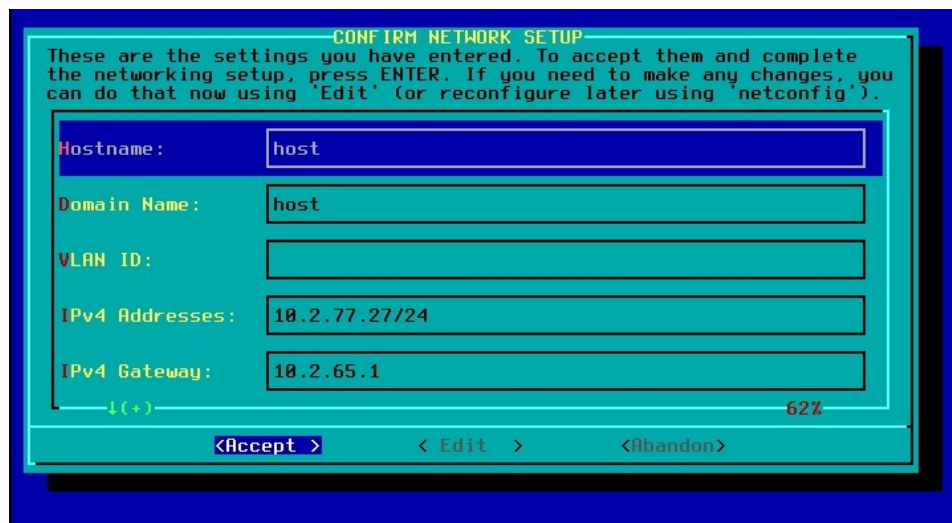
< OK >

<Cancel>



After a couple of options that focused more on boot details, and we didn't give it much importance because it is a minimal installation, we were finally able to configure the internet. To do this, the values given in the guide were placed in their respective spaces.





When the system was finally installed, everything worked except for the internet. In this case, we used the `ipconfig` command to find out if it was correctly configured. At that moment, we noticed that the mask was not correct and it was not a parameter that we were allowed to set. Given this situation, we decided to enter the internal configuration file to set the mask manually, for this we entered the file `X` within the `X` directory. Inside, we edited the file by placing `X` and then saved and exited.

```
GNU nano 6.0 rc.d/rc.inet1.conf Modified
# /etc/rc.d/rc.inet1.conf
#
# This file contains the configuration settings for network interfaces.
#
# If USE_DHCP[interfacel is set to "yes", this overrides any other settings.
# If you don't have an interface, leave the settings null ("").
#
# You can configure network interfaces other than eth0,eth1... by setting
# IFNAME[interfacel to the interface's name. If IFNAME[interfacel is unset
# or empty, it is assumed you're configuring eth<interface>.
#
# Several other parameters are available; the end of this file contains a
# comprehensive set of examples.
#
# Important note for IPv6 stateless auto configuration (SLAAC) users:
# From Slackware 15.0 onwards, you need to set USE_SLAAC[0]="yes" below.
#
# =====
# IPv4 config options for eth0:
IPADDRS[0]="10.2.77.27/24"
USE_DHCP[0]="""
NETMASK[0]="255.255.0.0"
# IPv6 config options for eth0:
IP6ADDRS[0]="""
USE_SLAAC[0]="""
USE_DHCP6[0]="""
# Generic options for eth0:
DHCP_HOSTNAME[0]="""
# IPv4 config options for eth1:
IPADDRS[1]="""
USE_DHCP[1]="""
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line  ^E Redo
```

Finally we can check that the internet works perfectly

```
root@chavarro:~# ping 10.2.65.1
PING 10.2.65.1 (10.2.65.1) 56(84) bytes of data.
64 bytes from 10.2.65.1: icmp_seq=1 ttl=64 time=7.92 ms
64 bytes from 10.2.65.1: icmp_seq=2 ttl=64 time=3.55 ms
64 bytes from 10.2.65.1: icmp_seq=3 ttl=64 time=4.11 ms
64 bytes from 10.2.65.1: icmp_seq=4 ttl=64 time=3.88 ms
64 bytes from 10.2.65.1: icmp_seq=5 ttl=64 time=29.4 ms
64 bytes from 10.2.65.1: icmp_seq=6 ttl=64 time=2.36 ms
64 bytes from 10.2.65.1: icmp_seq=7 ttl=64 time=2.96 ms
64 bytes from 10.2.65.1: icmp_seq=8 ttl=64 time=1.62 ms
```


NetBSD installation

Using the process described above, we create the machine on which we will install NetBSD. Once we have it ready, we start the machine, and we will see the first screen in the entire configuration.

```
NetBSD/amd64 10.0

This menu-driven tool is designed to help you install NetBSD to a hard disk,
or upgrade an existing NetBSD system, with a minimum of work.
In the following menus type the reference letter (a, b, c, ...) to select an
item, or type CTRL+N/CTRL+P to select the next/previous item.
The arrow keys and Page-up/Page-down may also work.
Activate the current selection from the menu by typing the enter key.

a: Installation messages in English
b: Installation auf Deutsch
c: Mensajes de instalacion en castellano
d: Messages d'installation en français
e: Komunikaty instalacyjne w jezyku polskim
```

In netBSD we are given the option to perform a minimal installation from the first moment. To do this, we select the respective option. However, we have to configure the operating system partition first, in this case the following steps we made it.

```
NetBSD/amd64 10.0

This menu-driven tool is designed to help you install NetBSD to a hard disk,
or upgrade an existing NetBSD system, with a minimum of work.
In the following menus type the reference letter (a, b, c, ...) to select an
item, or type CTRL+N/CTRL+P to select the next/previous item.
The arrow keys and Page-up/Page-down may also work.
Activate the current selection from the menu by typing the enter key.

Thank you for using NetBSD!

NetBSD-10.0 Install System

a: Install NetBSD to hard disk
b: Upgrade NetBSD on a hard disk
c: Re-install sets or install additional sets
d: Reboot the computer
e: Utility menu
f: Config menu
x: Exit Install System
```

On which disk do you want to install NetBSD?

Available disks

- >a: wd0 (3.0G, VBOX HARDDISK)
- b: Extended partitioning
- x: Exit

If you do not want to use the existing partitions, you can use a simple editor to set the sizes of the NetBSD partitions, or remove existing ones and apply the default partition sizes.

You will then be given the opportunity to change any of the partition details.

The NetBSD (or free) part of your disk (wd0) is 3072M.

A full installation requires at least 127M without X and at least 427M if the X sets are included.

What would you like to do?

- >a: Use existing disklabel partitions
- b: Set sizes of NetBSD partitions
- c: Use default partition sizes
- d: Manually define partitions
- x: Cancel

We are now going to install NetBSD on the disk wd0.

NetBSD requires a single partition in the disk's Master Boot Record (MBR) partition table, this is split further by the BSD disklabel. NetBSD can also access file systems in other MBR partitions.

If you select 'Use the entire disk' then the previous contents of the disk will be overwritten and a single MBR partition used to cover the entire disk. If you want to install more than one operating system then edit the MBR partition table and create a partition for NetBSD.

About 427M is enough for a basic installation, but you should allow extra for additional software and user files.

Allow at least 15G if you want to build NetBSD itself.

What would you like to do?

- a: Use existing MBR partitions
- >b: Edit the MBR partition table
- c: Use the entire disk
- d: Delete everything, use different partitions (not MBR)
- x: Cancel

You can now change the sizes for the system partitions. The default is to allocate all the space to the root file system. However, you may wish to have separate /usr (additional system files), /var (log files etc) or /home (users' home directories) file systems.

Free space will be added to the partition marked with a '+'.

Size (MB)	Filesystem
a: 1535	/
b: 1536	<swap>
c: 25%	/tmp (tmpfs)
d: 0	/usr
e: 0	/var

 g: Add a user defined partition
 h: Clone external partition(s)
 i: Change input units (sectors/cylinders/MB/GB)
 x: Go on. Free space 0 MB.

The Current MBR partition table of wd0 is shown below.
 Flags: (I)nstall here, (a)ctive, (d)efault boot.
 Select the partition you wish to change:

Total size of wd0 is 3072 MB, available: 0 MB

	Start(MB)	Size(MB)	Flag	Kind	Mount	Bootmenu
a:	0	1535	a	NetBSD		NetBSD
b:	1536	1536		Linux swap		

 d: Clone external partition(s)
 e: Change input units (sectors/cylinders/MB/GB)
 f: Cancel
 x: Partition table OK

With the disk already configured, there are only a couple of configurations left related to booting the system and configuring root. After all this, we can select the minimal installation and finish configuring the machine for now.

Would you like to install the normal set of bootblocks or serial bootblocks?
 Normal bootblocks use the BIOS console device as the console (usually the monitor and keyboard). Serial bootblocks use the first serial port as the console.

Selected bootblock: BIOS console

Bootblocks selection

a: Use BIOS console
 b: Use serial port com0
 c: Use serial port com1
 d: Use serial port com2
 e: Use serial port com3
 f: Set serial baud rate
 g: Use existing bootblocks
 x: Continue

The NetBSD distribution is broken into a collection of distribution sets. There are some basic sets that are needed by all installations and there are some other sets that are optional. You may choose to install a core set (Minimal installation), all of them (Full installation), or a custom group of sets (Custom installation).

Select your distribution

- a: Full installation
- b: Installation without X11
- c: Minimal installation
- d: Custom installation**
- x: Abandon installation

Your disk is now ready for installing the kernel and the distribution sets. As noted in your INSTALL notes, you have several options. For ftp or nfs, you must be connected to a network with access to the proper machines.

Sets selected 5, processed 0, Next set kern-GENERIC.

Install from

- a: CD-ROM / DVD / install image media**
- b: HTTP
- c: FTP
- d: NFS
- e: Floppy
- f: Unmounted fs
- g: Local directory
- h: Skip set
- i: Skip set group
- j: Abandon installation

Configure the additional items as needed.

- | | |
|---|--------------|
| a: Configure network | configure |
| b: Timezone | UTC |
| c: Root shell | /bin/sh |
| d: Change root password | password set |
| e: Enable installation of binary packages | install |
| f: Fetch and unpack pkgsrc | install |
| g: Enable sshd | NO |
| h: Enable ntpd | NO |
| i: Run ntpdate at boot | NO |
| j: Enable multicast DNS support | NO |
| k: Enable xdm | NO |
| l: Enable cgd | YES |
| m: Enable lvm | NO |
| n: Enable raidframe | YES |
| o: Add a user | |
| x: Finished configuring | |

Finally, and before rebooting, we configure the Internet by selecting the respective option given in the last menu shown. As with Slackware, we put in the given parameters the information from the guide to perform the correct configuration. In this case, at the end we perform the reboot and proceed to perform the respective tests to check the correct operation of the machines.

```
Chavarro# ping 8.8.8.8
PING dns.google (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=115 time=4.147175 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=7.694274 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=6.319518 ms
^C64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=9.602058 ms

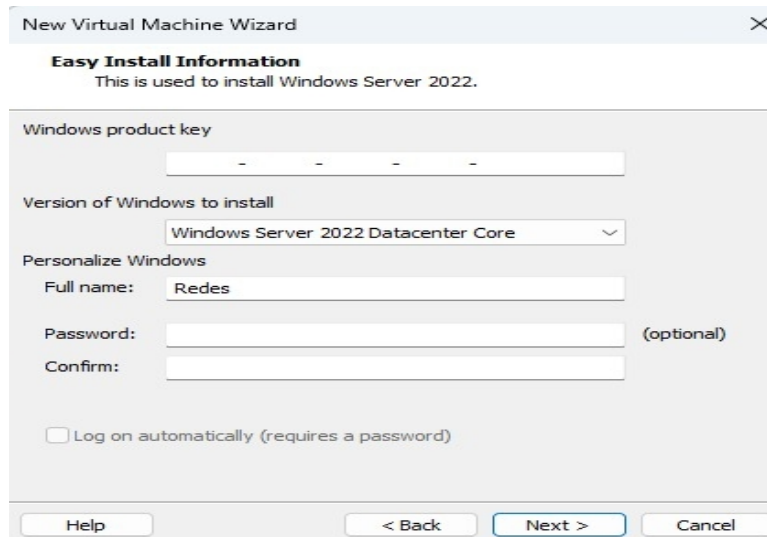
----dns.google PING Statistics----
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 4.147175/6.940756/9.602058/2.297850 ms
Chavarro# ping www.google.com
PING www.google.com (142.251.132.132): 56 data bytes
64 bytes from 142.251.132.132: icmp_seq=0 ttl=116 time=5.041982 ms
64 bytes from 142.251.132.132: icmp_seq=1 ttl=116 time=5.186413 ms
^C
----www.google.com PING Statistics----
2 packets transmitted, 2 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.041982/5.114198/5.186413/0.102128 ms
Chavarro# █
```

Installation summary

Comparing the installation and configuration of the two operating systems, we can find many similarities because although one is a Linux and the other a BSD, both are based on Unix. Therefore, all the internal and technical aspects are the same or very similar.

WINDOWS SERVER (NO GUI)

For this first machine that is created, we follow the machine creation process described in the section dedicated to VMware. Once the VM is configured, we can start it to boot with the installation after selecting the type of OS to be installed, in this case we select the Datacenter core version, which is the one that does not have a graphical interface.

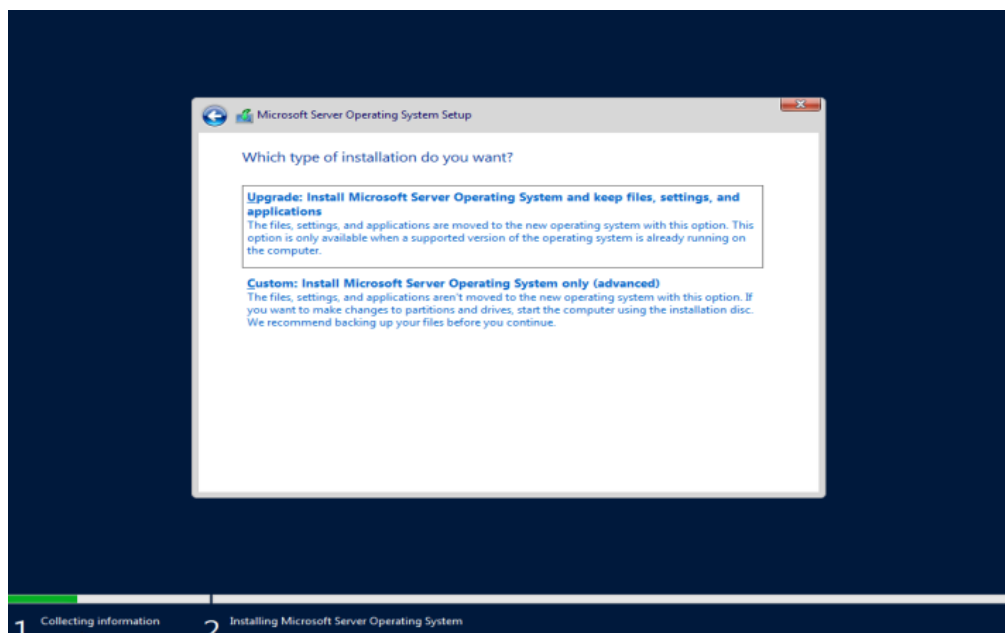


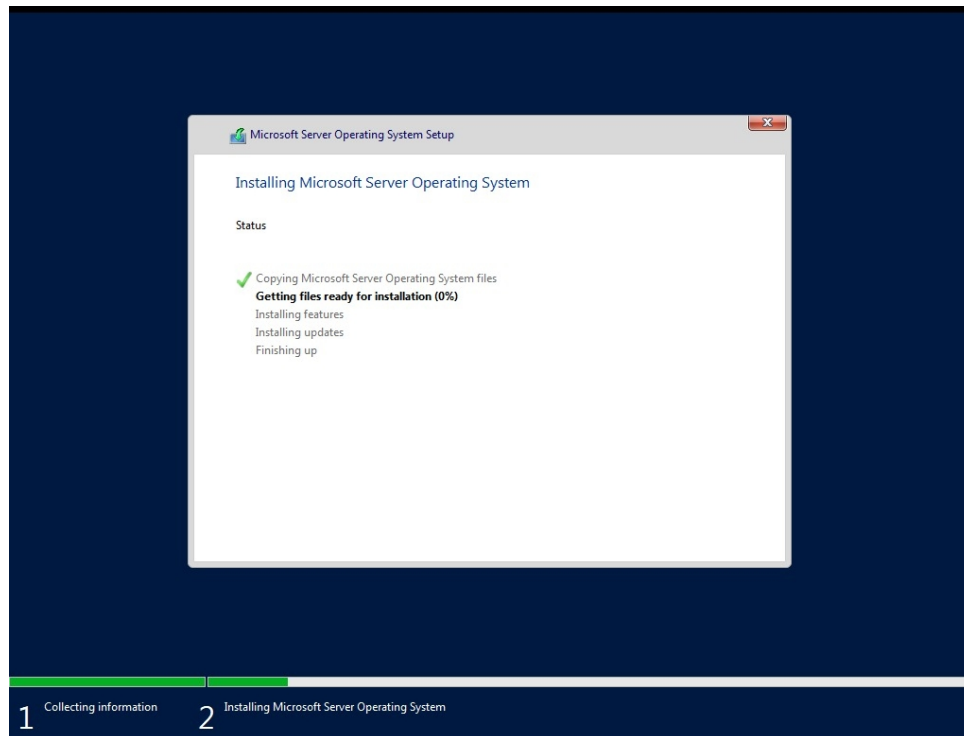
The screenshot shows the 'New Virtual Machine Wizard' window with the 'Easy Install Information' tab selected. The window title is 'New Virtual Machine Wizard'. Below the title bar, it says 'Easy Install Information' and 'This is used to install Windows Server 2022.' The main content area has the following fields and options:

- Windows product key:** A text box with four dashes '-'.
- Version of Windows to install:** A dropdown menu showing 'Windows Server 2022 Datacenter Core'.
- Personalize Windows:**
 - Full name:** A text box containing 'Redes'.
 - Password:** A text box, followed by '(optional)'.
 - Confirm:** A text box.
 - ☐ Log on automatically (requires a password)

At the bottom, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

First, a screen will appear indicating that the system is already running and the installation will begin as soon as the disk is accepted for use for the machine.





Once the system starts, put the root password and then we can proceed to configure the internet with the parameters given in the guide, for this we go to the netconfig system console to configure the internet.

```
=====
                        Network adapter settings
=====

NIC index:      1
Description:    Intel(R) 82574L Gigabit Network Connection
IP address:     169.254.59.70,
                fe80::a812:4611:7b95:3b46
Subnet mask:    255.255.0.0
DHCP enabled:   True

Default gateway:
Preferred DNS server: 10.2.65.60
Alternate DNS server: 10.2.65.61

  1) Set network adapter address
  2) Set DNS servers
  3) Clear DNS server settings

Enter selection (Blank=Cancel):
```



```
Administrator: C:\Windows\system32\cmd.exe

=====
                        Network adapter settings
=====

NIC index:      1
Description:    Intel(R) 82574L Gigabit Network Connection
IP address:     169.254.59.70,
                fe80::a812:4611:7b95:3b46
Subnet mask:    255.255.0.0
DHCP enabled:   True

Default gateway:
Preferred DNS server: 10.2.65.60
Alternate DNS server: 10.2.65.61

1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings

Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): s
Enter static IP address (Blank=Cancel): 10.2.77.30
Enter subnet mask (Blank=255.255.255.0): 255.255.0.0
Enter default gateway (Blank=Cancel): 10.2.65.1
Setting NIC to static IP...
Failed to release DHCP lease.
Result code: 83
Method name: ReleaseDHCPLease
(Press ENTER to continue): S
```

```
Administrator: C:\Windows\system32\cmd.exe

=====
                        Network adapter settings
=====

NIC index:      1
Description:    Intel(R) 82574L Gigabit Network Connection
IP address:     169.254.59.70,
                fe80::a812:4611:7b95:3b46
Subnet mask:    255.255.0.0
DHCP enabled:   True

Default gateway:
Preferred DNS server: 10.2.65.60
Alternate DNS server: 10.2.65.61

1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings

Enter selection (Blank=Cancel): 2
Enter new preferred DNS server (Blank=Cancel): 10.2.65.1
Enter alternate DNS server (Blank=None): 10.2.65.1
Successfully assigned DNS server(s).
(Press ENTER to continue):
```

Finally we performed the test and found that the internet works perfectly.

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\Administrator> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=18ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 18ms, Average = 18ms
Control-C
PS C:\Users\Administrator> ping www.google.com

Pinging www.google.com [142.250.78.164] with 32 bytes of data:
Reply from 142.250.78.164: bytes=32 time=6ms TTL=116
Reply from 142.250.78.164: bytes=32 time=3ms TTL=116

Ping statistics for 142.250.78.164:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 4ms
Control-C
PS C:\Users\Administrator> ping 10.2.65.1

Pinging 10.2.65.1 with 32 bytes of data:
Reply from 10.2.65.1: bytes=32 time=2ms TTL=64

Ping statistics for 10.2.65.1:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

WINDOWS SERVER (GUI)

For this one, we follow the same process of creating machines, but again we will get the option to change the type of OS and in this case the normal data center option will be set to have the Interfax.

New Virtual Machine Wizard

Easy Install Information
This is used to install Windows Server 2022.

Windows product key
| - - - -

Version of Windows to install
Windows Server 2022 Datacenter

Personalize Windows

Full name: Redes

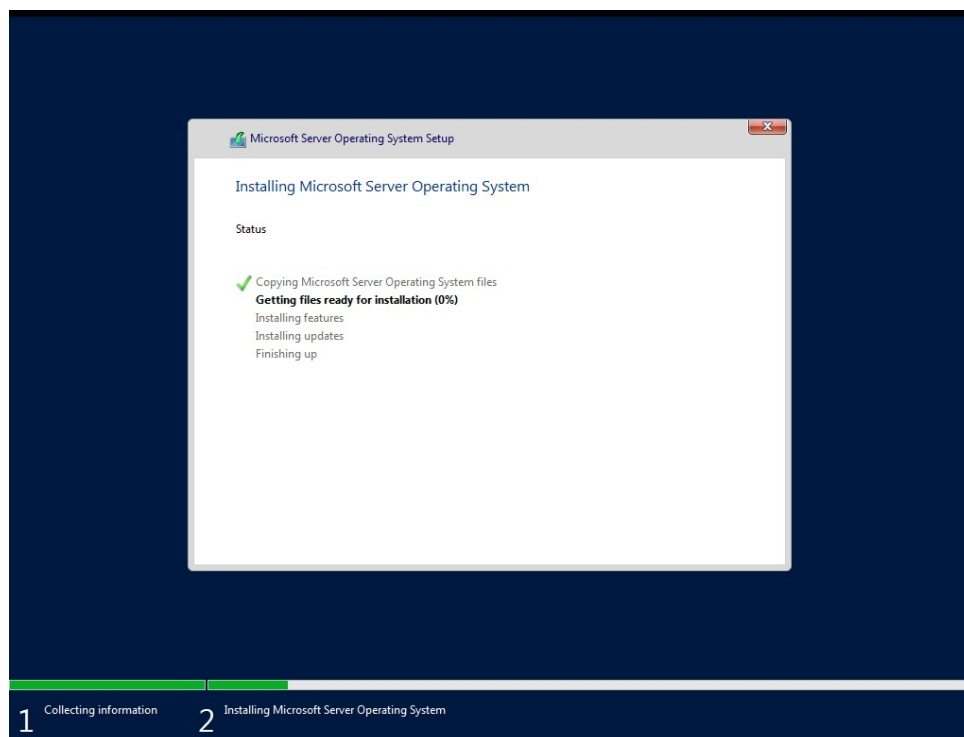
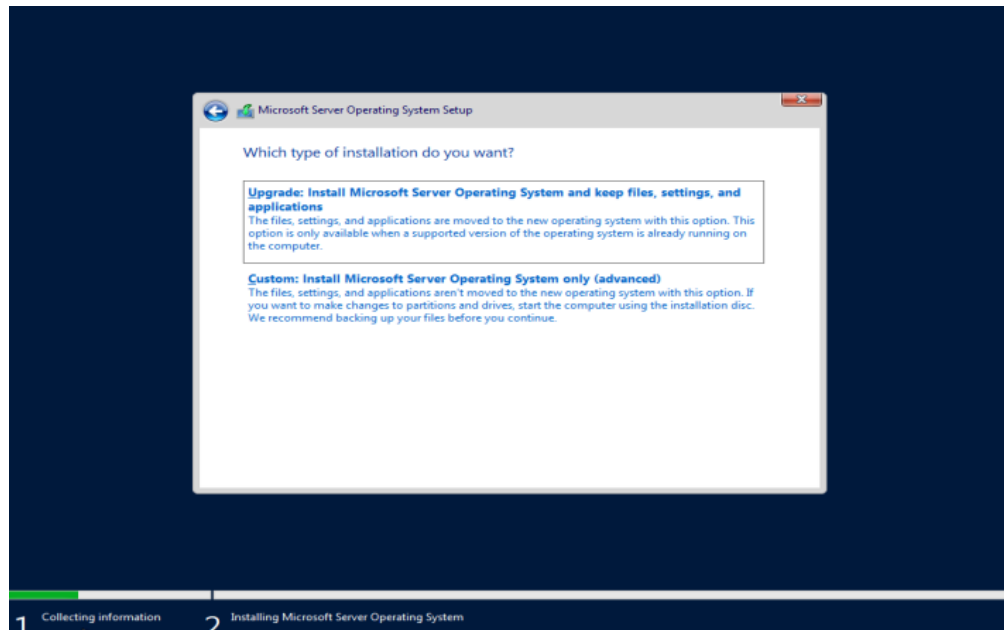
Password: (optional)

Confirm:

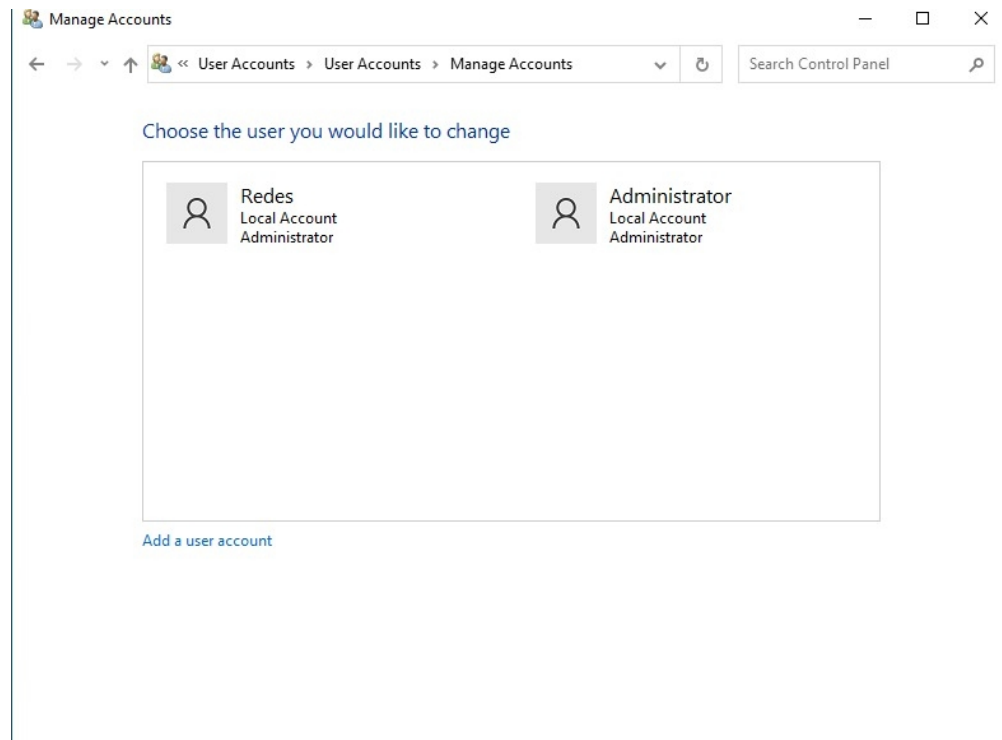
☐ Log on automatically (requires a password)

Help < Back Next > Cancel

Once the machine starts, you can see that it only asks you to validate the installation disk and then proceeds to install the system without any further configurations.



Once the system is initialized, you will see the following image. On the other hand, for the permissions issue, we go to the system settings and click on the users option. This allows us to create a user according to our needs.

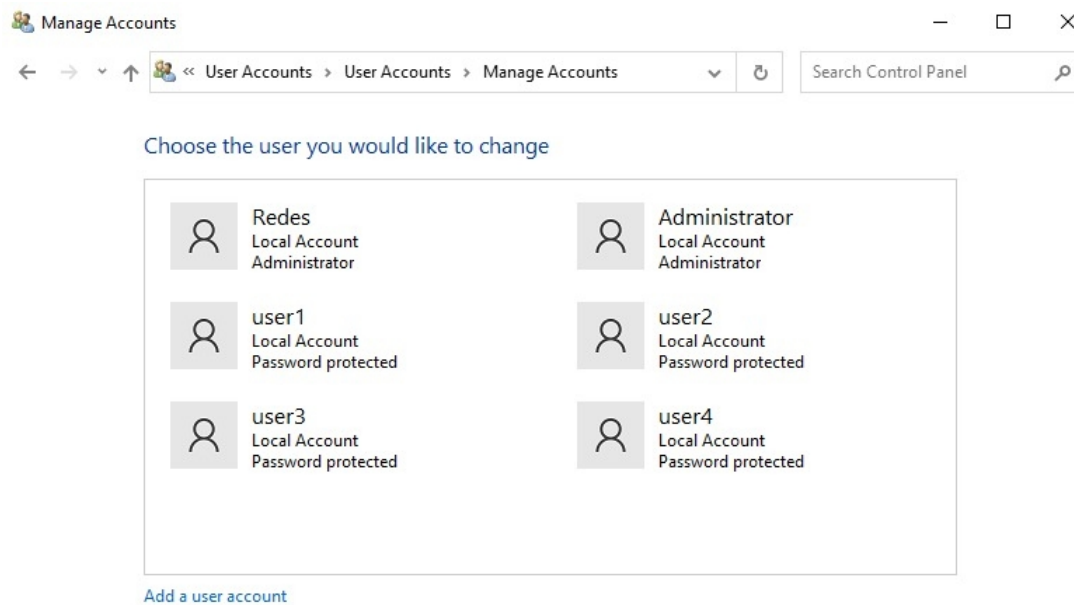


Add a user

Choose a password that will be easy for you to remember but hard for others to guess. If you forget, we'll show the hint.

Windows can't connect to the Internet right now. Check your Internet connection and try again later if you want to add a Microsoft account.

User name	user2
Password	••••••
Confirm password	••••••
Password hint	root ✕



Permissions in Windows Server are managed through Access Control Lists (ACLs), which define what actions users or groups can perform on files, folders, and other resources. NTFS file system permissions, such as read, write, or execute, control access to local files and directories. Additionally, share permissions manage access to networked folders. Tools like File Explorer, Active Directory, or PowerShell can be used to configure permissions. Features such as inheritance and group policies allow for efficient and centralized permission management across multiple resources.

The directory structure in Windows Server is hierarchical and includes key system folders. The C:\Windows directory contains essential operating system files, drivers, and services. The C:\Program Files directory is used for installed applications, while the C:\Users directory holds user profiles and data. Temporary files are stored in C:\Temp, and administrators rely on this structure to efficiently organize and manage system files and user data.

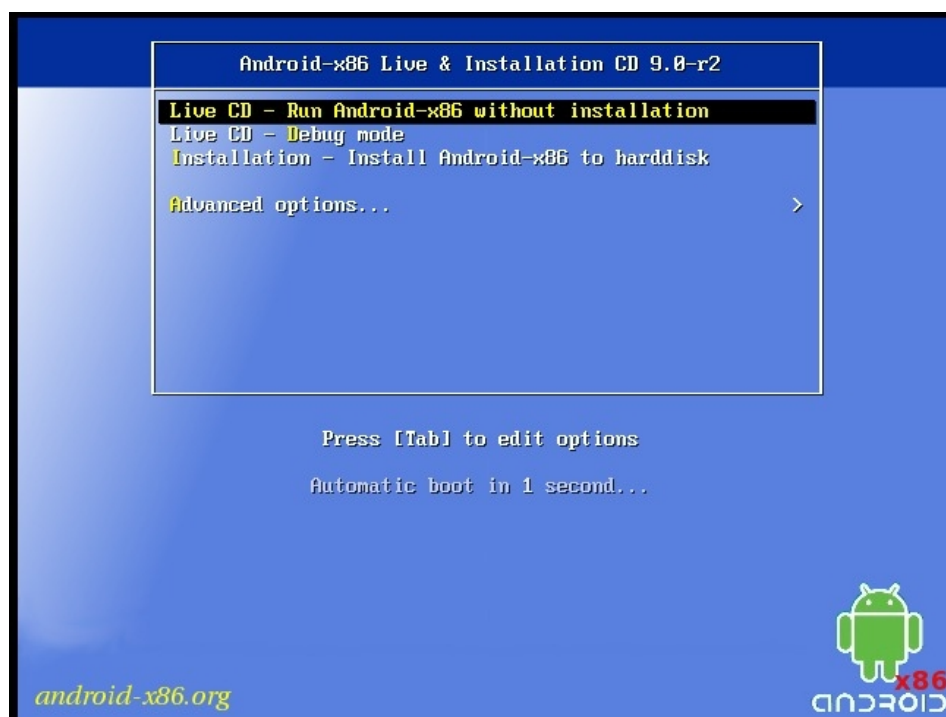
Windows Server logs are critical for monitoring and troubleshooting system performance, security events, and application issues. Logs are accessed via the Event Viewer, which is opened by typing eventvwr.msc in the Run dialog. The logs are organized into categories such as Application Logs, which record software events; Security Logs, which track login attempts and account changes; System Logs, which capture operating system and hardware events; and Setup Logs, which document

installation or configuration changes. Forwarded Events consolidate logs from other systems for centralized monitoring. Administrators can also use PowerShell commands like Get-EventLog and Get-WinEvent for advanced log management. Logs can be exported for analysis or archived for compliance and long-term monitoring.

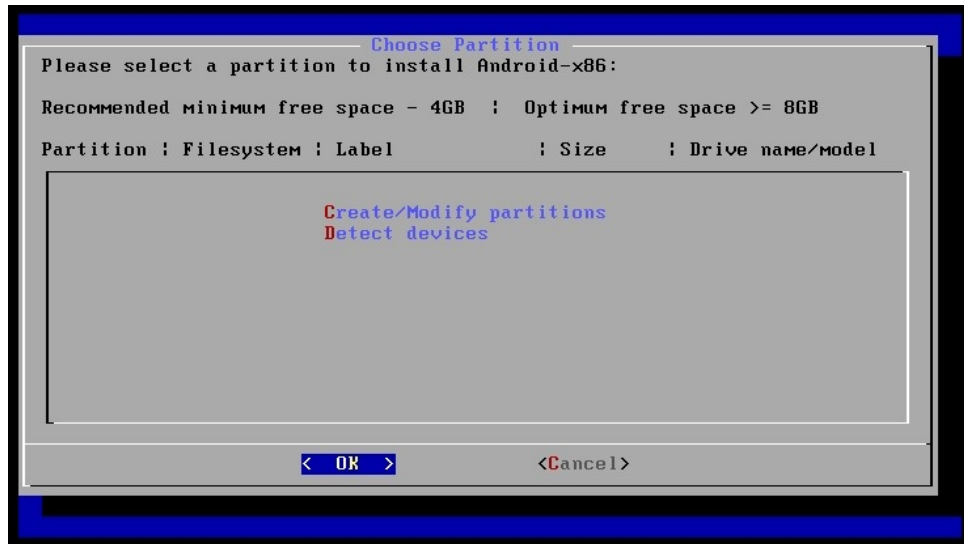
The Windows Registry is a hierarchical database that stores configuration settings and options for the operating system, installed software, hardware, and user preferences. It is essential for the proper functioning of Windows Server, as it governs system behavior and application configurations. The Registry can be edited using the Registry Editor (regedit), but caution is advised, as incorrect modifications can lead to system instability. It stores information such as file associations, user settings, device driver configurations, and installed software details. The Registry is divided into hives like HKEY_LOCAL_MACHINE (system-wide settings) and HKEY_CURRENT_USER (user-specific settings), making it a centralized and powerful configuration tool.

ANDROID

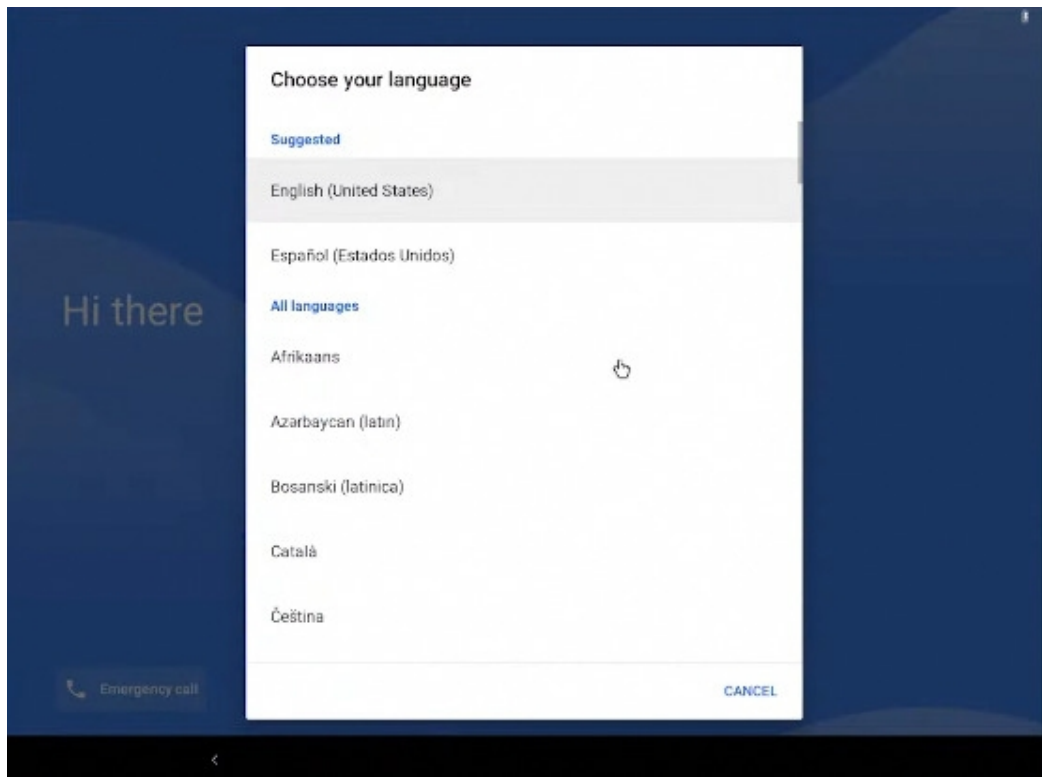
For this installation, the process already mentioned for creating the virtual machine was carried out again, so that only the OS installation remains. Initially, a screen appears where we choose what type of installation we are going to carry out.



Next, we need to partition the disk. On this one, we left a swap in case it was necessary and the rest for system storage.



Once we finished installing the system, we rebooted the machine to start the Android configuration, which would allow us to configure the network. However, in our case, only a console appeared without even possible commands, so we decided to replicate the installation at home and only got to the step of the startup screen.



COMMAND LINE

Shells

The shell is an interface that allows the user to interact with the operating system to execute commands and scripts. It acts as a bridge between the user and the system kernel. These shells are a set of functions and drivers that become the basis for handling files, scripts, processes, and more.

Supported shells of the installed OS's

Slackware: Bash, sh, zsh, ksh, tcsh

NetBSD: sh, csh, ksh, bash, zsh.

Windows: Command Prompt (cmd), PowerShell, WSL.

Differences from Unix shells and Windows shells.

In summary, shells in Unix systems stand out for their flexibility and orientation towards development and automation, while those in Windows focus on managing the Windows environment. Detailed differences are:

Shells in Unix systems:

- Structure: Text-based, ideal for combining commands using pipes.
- Scripting: Efficient, with support for cycles and functions.
- Use: Oriented towards data processing and advanced customization.

Shells in Windows:

- PowerShell: Works with objects instead of text, focused on system administration.
- Command Prompt: More limited and basic, used mainly for simple tasks.
- WSL: Integrates Linux functionalities within Windows.

CONCLUSIONS

To conclude this lab, we have been able to appreciate the importance of understanding and managing operating systems that can act as host OS on servers. In addition, knowing their internal structure and the way they interact with the hardware is essential, since this enables us to configure and manage networks or related services, which we will explore later in the course. On the other hand, the use of virtualization as a tool for running operating systems gives us a clear perspective on the management of virtual machines in various environments using specialized software, understanding both their structure and the operation of the files associated with each machine.

VIDEOS

For this lab we were asked to make a video corresponding to the theoretical part of it. The result is the following.

<https://youtu.be/hO9fqf2LqEM>

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